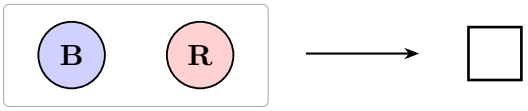
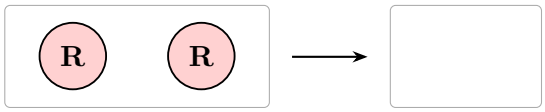
Use the same bag of three red counters and one blue. Draw twice in order, giving every counter the same chance each time. Return and mix after each draw.

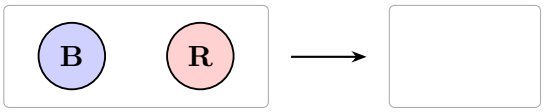


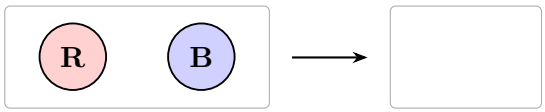
A sample rule: BR gives a square.

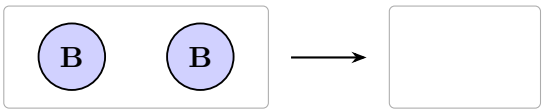
A rule gives each color pair a square, a circle, or no shape. A fair rule must sometimes give each shape, and give them equal chances when a shape comes out.

Problem 1: Design a fair rule for this bag. It must use only the colors and their order.

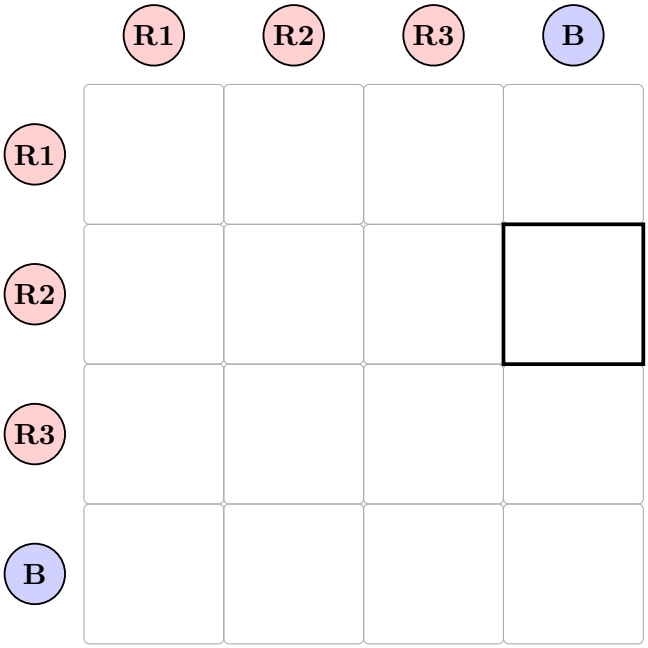


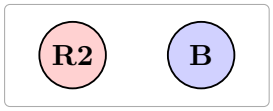






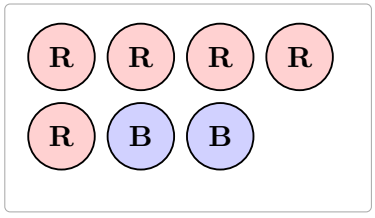
Problem 2: Check your rule using every equally likely pair of the labeled counters R1, R2, R3, B. Do the shape chances match? Explain.

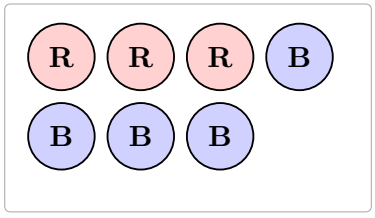




R2 then B is one labeled pair.
Its cell is outlined.

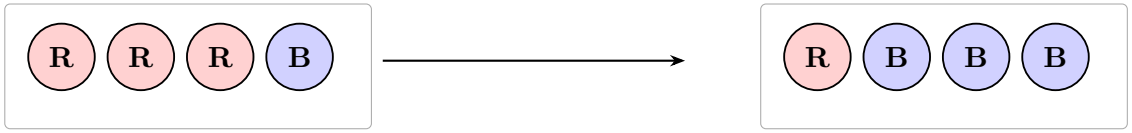
Problem 3: Keep your rule for these bags. Does it work for every fixed bag containing both colors, even if its contents are hidden? Explain your answer.



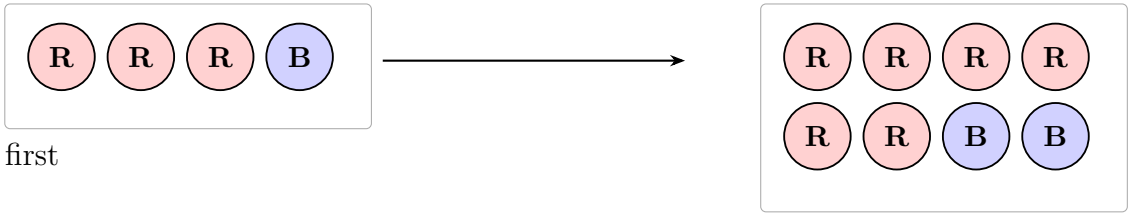


Problem 4: A different color-only rule must give a shape on every pair from the three-red, one-blue bag. Can it be fair? If skipping is allowed, what is the smallest number of the 16 labeled pairs it must skip?

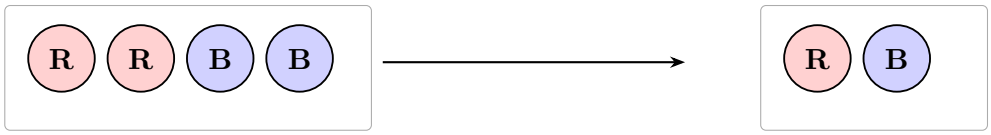
Problem 5: Use one bag for the first draw and another for the second. Decide exactly when your rule stays fair in the three cases below.



first second

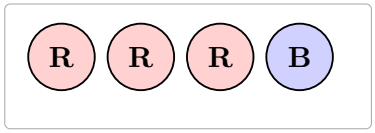


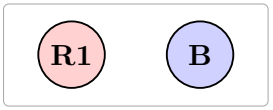
first second



first second

Problem 6: Return both counters only after drawing a pair. Then mix before the next pair. Does your rule still work with the three-red, one-blue bag? Explain using the labeled outcomes.





Problem 7: A friend says, "My rule is fair, so six pairs must give equal numbers of squares and circles." Another says, "At least one shape must appear." Decide which claims follow from fairness.